

```
MariaDB [(none)]> create database Industry;
```

Q7.24

```
MariaDB [Industry]> create table Employee(empNo varchar(50) PRIMARY KEY, fName
varchar(20), lName varchar(20), address varchar(40), DOB date, sex char(1) CHECK( SEX
IN('M','F')), position varchar(50) CHECK(POSITION IN('Manager','Team
Leader','Analyst','Software Developer')),deptNo varchar(10));
```

```
MariaDB [Industry]> create table Department(deptNo varchar(10) PRIMARY KEY,
deptName varchar(50),mgrEmpNo varchar(50), FOREIGN KEY(mgrEmpNo)REFERENCES
Employee(empNo));
```

```
MariaDB [Industry]> Alter table Employee add foreign key(deptNo)references
department(deptNo);
```

```
MariaDB [Industry]> create table Project(projNo varchar(50) primary key, projName
varchar(50),deptNo varchar(10),foreign key(deptNo) references
department(deptNo));
```

```
MariaDB [Industry]> create table WorksOn(empNo varchar(50),projNo varchar(50),
dateWorked date, hoursWorked decimal(4,2) check ( hoursWorked between 0 and 40));
```

```
MariaDB [industry]> insert into department values
```

```
-> ('D001','Administration',NULL),
-> ('D002','Human Resource',NULL),
-> ('D003','Finance',NULL),
-> ('D004','Sales',NULL),
-> ('D005','IT',NULL);
```

```
Query OK, 5 rows affected (0.031 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [industry]> insert into employee values
```

```
-> ('E001', 'John', 'Mensah', '12 Ring Road, Accra', '1995-04-15', 'M',
'Manager', 'D001'),
-> ('E002', 'Ama', 'Owusu', '45 High Street, Kumasi', '1998-09-23', 'F', 'Team
Leader', 'D002'),
-> ('E003', 'Kwame', 'Asante', '18 Liberation Road, Accra', '1992-11-08', 'M',
'Analyst', 'D003'),
-> ('E004', 'Abena', 'Boateng', '7 Oxford Street, Takoradi', '1997-06-30', 'F',
'Software Developer', 'D004'),
-> ('E005', 'Kojo', 'Antwi', '25 Adenta Road, Accra', '1990-01-17', 'M',
'Manager', 'D005');
```

```
Query OK, 5 rows affected (0.029 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [industry]> update department set mgrEmpNo = 'E001'where deptNo = 'D001';
```

```
Query OK, 1 row affected (0.034 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [industry]> update department set mgrEmpNo = 'E002'where deptNo = 'D002';
```

Query OK, 1 row affected (0.027 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [industry]> update department set mgrEmpNo = 'E003'where deptNo = 'D003';
Query OK, 1 row affected (0.026 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [industry]> update department set mgrEmpNo = 'E004'where deptNo = 'D004';
Query OK, 1 row affected (0.027 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [industry]> update department set mgrEmpNo = 'E005'where deptNo = 'D005';
Query OK, 1 row affected (0.027 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [industry]> insert into project values
-> ('P001', 'Payroll System', 'D001'),
-> ('P002', 'Employee Management System', 'D002'),
-> ('P003', 'Budget Analysis', 'D003'),
-> ('P004', 'Sales Tracking System', 'D004'),
-> ('P005', 'Network Upgrade', 'D005');
Query OK, 5 rows affected (0.028 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [industry]> insert into workson values
-> ('E001', 'P001', '2026-06-01', 8.00),
-> ('E001', 'P002', '2026-06-02', 6.50),
-> ('E002', 'P002', '2026-06-01', 7.00),
-> ('E002', 'P003', '2026-06-03', 8.00),
-> ('E003', 'P003', '2026-06-02', 5.50),
-> ('E003', 'P004', '2026-06-04', 7.50),
-> ('E004', 'P004', '2026-06-01', 8.00),
-> ('E004', 'P005', '2026-06-05', 6.00),
-> ('E005', 'P001', '2026-06-03', 7.00),
-> ('E005', 'P005', '2026-06-06', 8.00);
Query OK, 10 rows affected (0.029 sec)
Records: 10 Duplicates: 0 Warnings: 0

Q7.25

MariaDB [Industry]> create view FemaleManagedProject as select projName, projNo
from project,department, employee where project.deptNo = department.deptNo and
department.mgrEmpNo = employee.empNo and employee.sex = 'F' order by projNo;
MariaDB [industry]> select * from FemaleManagedProject;

projName	projNo
Employee Management System	P002
Sales Tracking System	P004

```
+-----+-----+
2 rows in set (0.039 sec)
```

Q7.26

```
MariaDB [Industry]> create view EmployeeView as select
employee.empNo,fName,lName,projName, hoursWorked from employee,project,workson
where project.projNo = workson.projNo and employee.empNo = workson.empNo;
```

```
MariaDB [industry]> select * from EmployeeView;
```

empNo	fName	lName	projName	hoursWorked
E001	John	Mensah	Payroll System	8.00
E001	John	Mensah	Employee Management System	6.50
E002	Ama	Owusu	Employee Management System	7.00
E002	Ama	Owusu	Budget Analysis	8.00
E003	Kwame	Asante	Budget Analysis	5.50
E003	Kwame	Asante	Sales Tracking System	7.50
E004	Abena	Boateng	Sales Tracking System	8.00
E004	Abena	Boateng	Network Upgrade	6.00
E005	Kojo	Antwi	Payroll System	7.00
E005	Kojo	Antwi	Network Upgrade	8.00

```
10 rows in set (0.032 sec)
```

Q7.27

```
MariaDB [industry]> create view EmpProject(empNo,projNo,totalHours) as select
w.empNo,w.projNo, sum(hoursworked) from employee e,project p, workson w where
e.empNo = w.projNo and p.projNo = w.projNo group by w.empNo, w.projNo;
Query OK, 0 rows affected (0.038 sec)
```

```
a. MariaDB [industry]> select * from EmpProject;
```

```
MariaDB [industry]> select * from EmpProject;
```

empNo	projNo	totalHours
E001	P001	8.00
E001	P002	6.50
E002	P002	7.00
E002	P003	8.00
E003	P003	5.50
E003	P004	7.50
E004	P004	8.00
E004	P005	6.00
E005	P001	7.00
E005	P005	8.00

```
10 rows in set (0.012 sec)
```

b.MariaDB [industry]> select projNo from EmpProject where projNo = 'SCCS';
Note: Since my database uses project numbers such as P001 instead of SCCS, I replaced 'SCCS' with 'P001' in the query to match my data.

MariaDB [industry]> select projNo from EmpProject where projNo = 'P001';

```
+-----+
| projNo |
+-----+
| P001   |
| P001   |
+-----+
```

2 rows in set (0.010 sec)

c.MariaDB [industry]> select count(projNo) from EmpProject where empNo = 'E001';
Note: I changed 'E1' to 'E001' because my database uses employee numbers in the format E001, E002,

```
+-----+
| count(projNo) |
+-----+
|              2 |
+-----+
```

1 row in set (0.031 sec)

d.MariaDB [industry]> select empNo, totalHours from EmpProject group by empNo;

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E001  |          8.00 |
| E002  |          7.00 |
| E003  |          5.50 |
| E004  |          8.00 |
| E005  |          7.00 |
+-----+-----+
```

5 rows in set (0.027 sec)

Q7.28

The materialized view is maintained by updating it whenever a row in the Part table is inserted, updated, or deleted. A part is added to the view if its cost becomes greater than £1000 and removed if it no longer satisfies the condition.

The materialized view can be maintained without accessing the base table when a newly inserted or updated row clearly satisfies the condition $\text{partCost} > 1000$. However, when a qualifying row is updated or deleted, the base table must be checked to determine whether another row with the same partNo still has a cost greater than £1000.

